

Neural correlates of impulsive-violent behavior: an event-related potential study

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Impulsive-violent offenders are often less capable of controlling their behavior. A cued Go/No-go task was used to test the hypothesis that impulsive-violent behavior reflects a deficit in the ability to inhibit prepotent responses and could be reflected in the measurement of event-related potentials. Results showed that the amplitudes of the N2 component at Fz reflected different degrees of inhibition in impulsive-violent offenders compared with matched

controls. The N2 amplitude (No-go minus Go) was significantly lower in the impulsive-violent offenders than in matched controls. The amplitude of N2 increased when effort was required to withhold the Go intention. A smaller N2 amplitude was seen in offenders, suggesting difficulties with inhibition of prepotent behavior. *NeuroReport* 16:1213–1216 © 2005 Lippincott Williams & Wilkins.

Key words: Event-related potentials; Go/No-go task; Impulsive-violent behavior; N2

INTRODUCTION

Many species have the ability to inhibit aggressive attacks when a conspecific displays submission cues. Humans possess a similar capacity in the context of distress cues. The violent inhibition mechanism could initiate withdrawal responses to terminate aggressive behavior [1] and the behavioral inhibition system may suppress inappropriate behavior [2–4], implying that behavioral inhibition is associated with the control of violent behavior. Ability to control motor responses is an effective measure of this process. Previously, Go/No-go tasks have been used to probe action control in aggressive adolescents who made more errors on No-go trials, suggesting problems with action inhibition [5].

Event-related potentials (ERPs) have been used to investigate the electrophysiological correlates of motor inhibition in Go/No-go tasks. Two main differences between Go and No-go trials are seen; a negative deflection in the ERPs evoked in No-go trials compared with Go trials in the 200–400 ms time period (N2 component) at the frontocentral location (Fz) and a positive wave peaking between 300 and 600 ms latency (P3 component) at Fz and Cz [6]. The amplitude of the No-go P3 is enhanced compared with the Go P3 and is more anterior in its distribution [7,8]. The finding that the P3, but not the N2, is associated with response inhibition in response priming studies supports the motor inhibition interpretation of the Go/No-go effect [9].

N2 amplitude effects are also seen in No-go trials with a higher N2 signal when under response time pressure [6,10–12]. Go and No-go trials differ in their required preparation for motor execution, so the difference in the N2 signals between Go and No-go trials may reflect the readiness to execute a motor response, or study participants may be able

to link their responses to the cues without effort, suggesting that differences may reflect differing levels of attentional resources. To investigate the influence of the state of motor programming on N2 amplitude, cues carrying different amounts of information as to the response direction can be used [11]. Different priming conditions modulate the N2, but not the P3, on No-go trials, so the N2 might reflect the inhibition of an inappropriate tendency to respond. Consistent with this, the amplitude of the No-go N2 in participants who made few errors was larger than that in participants who made many errors [12]. Differences between groups in the amplitude and latency of the No-go N2 suggest that inhibition of an intended behavior correlates more with the No-go N2 than the No-go P3.

The inhibition hypothesis is also corroborated by animal studies [13,14]. Like humans, the N2 was related to the No-go trials and the origin of the N2 might be neurons in the caudal part of the dorsal bank of the principal sulcus, consistent with human ERP experiments. Human patient and neuroimaging studies provide strong evidence for the role of the prefrontal cortex in the inhibition mechanism [15]. Patients with lesions of the orbitofrontal cortex show impaired performance on the Go/No-go task and functional magnetic resonance imaging studies have shown activation of medial and inferior parts of the left frontal cortex [16], supporting a crucial role for a higher-level inhibition mechanism mediated by the frontal cortex.

The aims of the present study were to (1) clarify whether the N2, P3 or both ERP components are indicators of the inhibition mechanism and (2) test whether impulsive-violent offenders' and normal participants' inhibition mechanisms could be differentiated at behavioral and electrophysiological levels.

MATERIALS AND METHODS

Seventeen male impulsive-violent offenders (mean age 27.5 ± 4.4 years) and seventeen controls who had also committed offenses but not of the impulsive-violent type (mean age 29.1 ± 3.8 years) matched for age, gender and education participated in behavioral Experiment 1. All offenders were recruited from the Taiwan Taipei prison and had committed more than one homicide (unpremeditated), using their hands, knives or sticks, and had been sentenced to life imprisonment or had served more than 15 years' imprisonment. Participants did not include those charged with robbery, murder in the heat of passion, rape, homicide while intoxicated or while under the influence of drugs. Eight right-handed male volunteers (mean age 18.7 ± 0.9 years) participated in ERP Experiment 2. Three volunteer male violent offenders (mean age 33.3 ± 3.2 years) who had been released on parole and three matched controls (mean age 32 ± 2.6 years) participated in ERP Experiment 3. To be included, they had to have served 15 years' imprisonment and had to be below 40 years of age to exclude age-related ERP changes. This limited the number of volunteers available for this experiment. All participants had normal or corrected-to-normal vision, gave written, informed consent before participating in the study and all experiments were approved by the Ministry of Justice and the ethics committee of the Taipei Prison, Taiwan.

Two tasks were used: (1) a central cued Go/No-go task and (2) a peripheral cued Go/No-go task. In the central cued Go/No-go task, a trial consisted of a cue (specific No-go prime [], nonspecific Go prime < >, specific Go prime <] or [>) and a target frame (No-go, left or right). The specific No-go prime was always associated with the No-go target. The nonspecific Go prime could be followed by any of the targets. The two specific Go primes were associated with the No-go target and a specific Go target (left or right). All stimuli were presented in the center of the screen and were white against a black background. Arrowheads were constructed as equilateral triangles with an edge length of 18 mm. The diameter of the circle was 18 mm. Trials began with a 500 ms central fixation cross. The duration of the cues was 100 ms, and that of the targets was 200 ms. The stimulus onset asynchrony between the cue and target frame was 350 ms. The peripheral cued Go/No-go task consisted of a flanker cue (nonspecific Go prime ■, specific Go prime ◀ or ▶) and a target (No-go, left or right). As stimulus onset asynchronies between flanker and target stimuli decreased, the visual context effect was enhanced [17,18]; flanker onset preceded the onset of the target by 100 ms. The target was presented in addition to the flanker stimuli and stayed at the center of the screen for 50 ms. All three stimuli were removed simultaneously. Participants had to respond with their index finger when a left arrowhead was presented, their middle finger when a right arrowhead was presented and withhold responses when a circle was presented. Each task consisted of one practice block and seven experimental blocks. Forty trials per block were carried out in task 1 and 45 trials per block in task 2.

Experiment 1 took place in a sound-attenuated room in prison and participants faced an LCD screen at a distance of 60 cm. In Experiments 2 and 3, participants were seated in a shielded room facing a screen at a distance of 100 cm. The electroencephalogram (EEG) signal was recorded using a 64-electrode arrangement according to the International

10–20 system and referred to the left mastoid. The vertical and horizontal electrooculogram (EOG) was recorded from sites above and below the left eye and 1 cm external to the outer canthus of each eye. A ground electrode was placed on the forehead. Electrode impedance was kept below 5 k Ω . EEG and EOG signals were amplified with a 500 s time constant. The amplifier bandpass was 0.05–50 Hz. Epochs started 100 ms before target onset and lasted 780 ms. Trials with eye blinks (EOG greater than $\pm 50 \mu\text{V}$) or with a movement artifact (EEG greater than $\pm 100 \mu\text{V}$) were rejected as were epochs with amplifier saturation artifacts or incorrect responses. ERPs were aligned to a prestimulus baseline of –100 to 0 ms before target onset. Averages were computed separately for each task, participant, block and stimulus category for the midline electrode sites (Fz, Cz, Pz and Oz). These averages were group-averaged over blocks (individual grand averages) and participants (across-participants grand averages).

RESULTS

Data were analyzed by means of repeated-measures ANOVA, unless stated.

Behavioral data (Experiment 1): In the central cued Go/No-go task, no significant difference was observed between the RTs of the two groups across the summed Go response conditions [$F(1,32)=5.038$, $p>0.01$]. The mean RT to Go targets was shorter for specific Go primes (410 ± 47.5 ms) than for nonspecific Go primes [419 ± 43.4 ms; $F(1,32)=7.706$, $p<0.01$]. In the peripheral cued task, the mean RT of the impulsive-violent offenders (417 ± 46.9 ms) was slower than that of controls [378 ± 25.3 ms; $F(1,32)=12.1$, $p<0.01$]. A significant effect for the visual context [congruent, neutral and incongruent, $F(1,32)=114.616$, $p<0.01$] was observed. No significant difference between the two groups was observed in the Go response condition. In both tasks, no significant effects of specific priming were observed on the proportion of hits, false alarms, or misses, nor were any differences in the proportion of correct rejections or false alarms observed in response to No-go targets in the different priming conditions.

Event-related potential data (Experiment 2): (1) *Effects of priming on inhibition (No-go targets)* In the central cued Go/No-go task, the centrofrontal maximal N2 was largest for the specific Go prime conditions (Fig. 1) and was completely absent for the No-go targets in the specific No-go prime condition. A significant main effect for prime was seen for the mean N2 amplitudes [$F(1,6)=6.415$, $p<0.05$]. Furthermore, a significant difference between the ERPs evoked in the specific Go prime condition and specific No-go prime condition was found at Fz in the 230–310 ms interval (Student's t -test, $t=-1.972$, $p<0.05$). In the peripheral cued task, the centrofrontal maximal N2 was largest for the specific Go prime condition (Fig. 2). This N2 component was absent for No-go targets in the nonspecific prime (neutral flanker stimuli) condition. A significant main effect for prime for the mean N2 amplitudes was seen [$F(1,6)=9.372$, $p<0.05$]. A significant difference between the ERPs evoked in the specific Go prime condition and nonspecific prime condition was found at Fz in the interval 200–270 ms (Student's t -test, $t=4.896$, $p<0.05$). No significant differences

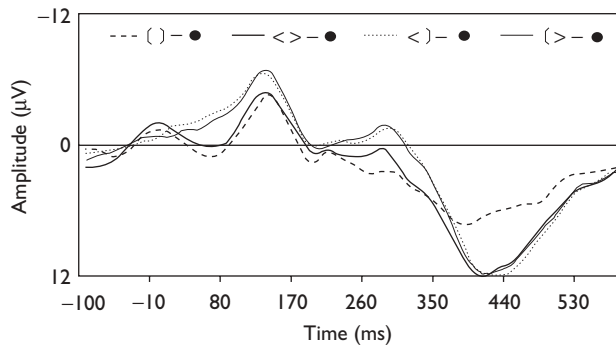


Fig. 1. Grand-average event-related potentials evoked by No-go targets at Fz, N2 in the specific Go prime condition was larger than in other conditions. The No-go targets were preceded by a specific No-go prime ([]), a nonspecific Go prime (< >), or one of the two specific Go primes (< >) or (> <).

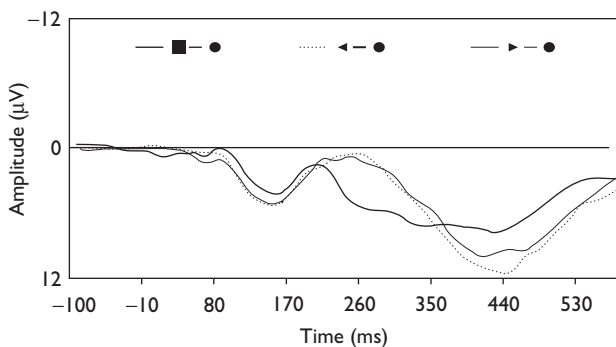


Fig. 2. Grand-average event-related potentials evoked by No-go targets at Fz. The No-go targets were preceded by a nonspecific Go prime, or one of the two specific Go primes. When the nonspecific prime displayed in the No-go trials, N2 component was absent. The N2 component was largest in the specific Go prime condition.

in the amplitudes of the ERPs evoked by No-go targets in the different prime conditions in the P3 interval were observed in the two-cued Go/No-Go task.

(2) *Effects of target type given a specific Go prime* In the central cued task, the amplitude of the ERPs to No-go targets compared to that of the ERPs to Go targets was more negative at Fz in the interval 230–310 ms. A significant main effect for target types was seen for the mean N2 amplitudes [$F(1,6)=29.891, p<0.05$]. This effect replicates the previously observed N2 Go/No-go effect. A larger P3 wave was seen for No-go targets than for Go targets at Fz in the interval 380–460 ms. A significant effect of target type was seen for P3 amplitudes from No-go trials [$F(1,6)=14.827, p<0.05$].

In the peripheral cued task, the amplitude of the ERPs to No-go targets (specific Go prime trials) compared to that of the ERPs to Go targets (congruent visual context) was more negative at Fz in the interval 200–270 ms. The mean N2 amplitudes revealed a significant main effect for target types [$F(1,6)=18.821, p<0.05$]. The amplitude of the ERPs to incongruent visual context was more negative at Fz in the interval 270–340 ms. A significant main effect for visual context for the mean N2 amplitudes was seen [$F(1,6)=14.365, p<0.05$]. No statistically significant differences in the amplitudes of the ERPs evoked by No-go

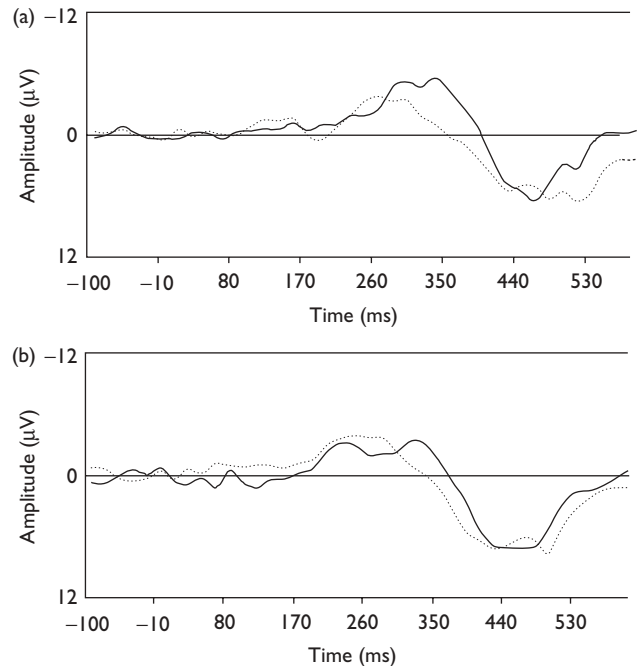


Fig. 3. Difference waves (No-go minus Go preceded by specific Go primes) for the offender group (dashed line) were smaller than for the matched controls (solid line) at Fz in both the central cued task (a) and peripheral cued task (b).

targets were observed in the difference prime conditions for the P3 interval (ca. 340–410 ms).

Event-related potential data (Experiment 3): To investigate N2 and P3 differences, difference wave shapes were computed for the ERPs of No-go and Go targets preceded by specific Go prime (No-go minus Go). The waves for impulsive-violent offenders were compared with those of the matched controls. In the central cued Go/No-go task, a significant group main effect [$F(1,4)=9.85, p<0.05$] confirmed that the N2 at Fz was larger for matched controls ($3.56 \mu\text{V}$) than for the offender group ($0.12 \mu\text{V}$) (Fig. 3a). The same comparison for the peripheral cued Go/No-go task showed a significant group main effect [$F(1,4)=35.71, p<0.05$] with a larger N2 at Fz for the control group ($1.88 \mu\text{V}$) than for the offender group ($0.31 \mu\text{V}$) (Fig. 3b). No significant difference in the amplitude of the P3 waves was observed between the groups in either task.

Difference waves for incongruent visual context minus congruent visual context (Go target preceded by specific Go prime) were significantly larger in the control group ($3.58 \mu\text{V}$) than in the offenders ($0.87 \mu\text{V}$) [$F(1,4)=11.01, p<0.05$] (Fig. 4). No significant difference was observed between groups in the amplitude of P3 to the incongruent context minus the congruent context. The P3 showed no consistent amplitude difference between the two groups.

DISCUSSION

The results from Experiment 1 lengthened reaction times to Go targets preceded by a nonspecific Go prime compared with when the Go target was preceded by a specific Go prime, and larger flanker compatibility effects in Go trials, suggest that the cued Go/No-go task successfully elicited

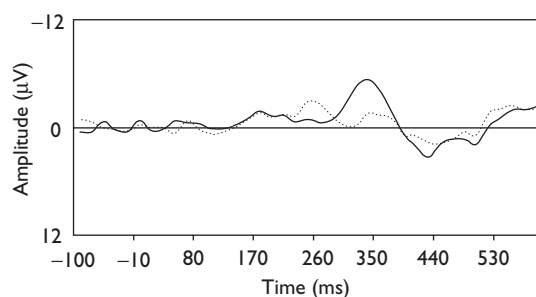


Fig. 4. Difference waves (Go targets preceded by incongruent Go primes minus Go target preceded by congruent Go primes) for the offender group (dashed line) were smaller than for the matched controls (solid line) at Fz in the peripheral cued task.

the inhibition mechanism. However, it was not possible to distinguish impulsive-violent offenders from controls with these measures.

In Experiment 2, established N2 Go/No-go ERP effects were replicated, with the centrofrontal N2 component enhanced when a Go response had to be withheld [7,14], consistent with the involvement of the frontal lobe in the inhibition mechanism. The N2 response to the No-go stimulus increased in amplitude when a greater effort was required to withhold the Go response. Because it has been argued that when comparing Go and No-go trials, response inhibition is confounded by motor potentials [10,11], we compared ERPs evoked by No-go targets in terms of their different primes. We found that only the N2 was influenced by this prime. A distinct N2 appeared most prominently at the vertex in the incongruent visual context condition of the Go trials and in the specific priming condition of the No-go trials. In the latency range from 200 to 400 ms, the appearance of the N2 was not caused by either cue or target alone, but by a combination of the two.

Our results differ from previous results in which No-go trial response priming influenced the P3 but not the N2 [9]. The absence of variation of the P3 amplitude as a function of erroneous response priming in their study suggested that the traditional P3 Go/No-go effect is associated with overlaying motor potentials rather than response inhibition [11]. In contrast, the N2 effects seem to be modulated by response inhibition processes, suggesting that N2 reflects the activity of a response inhibition system of the brain and is consistent with the neural systems, not only initiating behavior but also suppressing them actively.

Experiment 3 showed that the N2 to the No-go stimulus or incongruent cue-target combinations was larger in the control group than in the offender group in both cued Go and No-go tasks. The increased N2 amplitude in response to No-go stimuli after Go cues might be related to increased efforts to activate the response inhibition mechanism and to interrupt preparations for response execution [18]. We suggest that the impulsive-violent offenders may not have sufficient neural resources to activate the inhibition mechanism. We did not find significant difference in the P3 component between two participant groups. Our results also revealed the existence of different inhibitory processes

supported by the frontal lobe, as response patterns differed between the impulsive-violent group and matched controls.

CONCLUSION

Results here suggest that the N2, but not the P3, is associated with response inhibition, with the response priming effect on the N2 amplitude reflecting the recruitment of neural activity that serves to suppress the execution of otherwise inappropriate responses. The N2 amplitude in the impulsive-violent group was weaker than that of the matched controls, providing convergent evidence supporting the N2 hypothesis as the neuropsychological basis of the inhibition mechanism.

REFERENCES

- Blair RJR. A cognitive development approach to morality: Investigating the psychopath. *Cognition* 1995; **57**:1–29.
- Gray JA. A critique of Eysenck's theory of personality. In: Eysenck HJ (ed.). *A Model of Personality*. New York: Springer Press; 1981. pp. 246–276.
- Gray JA. Brain systems that mediate both emotion and cognition. *CognEmotion* 1990; **4**:269–288.
- Gray JA. Personality dimensions and emotion systems. In: Ekman P, Davidson RJ (eds). *The Nature of Emotion: Fundamental Questions*. New York: Oxford University Press; 1994. pp. 329–331.
- LeMarquand DG, Pihl RO, Young SN, Tremblay RE, Seguin JR, Palmour RM *et al.* Tryptophan depletion, executive functions, and disinhibition in aggressive, adolescent males. *Neuropsychopharmacology* 1998; **19**:333–341.
- Jodo E, Kayama Y. Relation of a negative ERP component to response inhibition in a Go/Nogo task. *Electroencephalogr Clin Neurophysiol* 1992; **82**:477–482.
- Pfefferbaum A, Ford JM, Weller BJ, Kopell BS. ERPs to response production and inhibition. *Electroencephalogr Clin Neurophysiol* 1985; **60**:423–434.
- Roberts LE, Rau H, Lutzenberger W, Birbaumer N. Mapping P300 waves onto inhibition: Go/Nogo discrimination. *Electroencephalogr Clin Neurophysiol* 1994; **92**:44–55.
- Bruin KJ, Wijers AA, Staveren ASJ. Response priming in a go/nogo task: do we have to explain the go/nogo N2 effect in terms of response activation instead of inhibition? *Clin Neurophysiol* 2001; **112**:1660–1671.
- Kok A. Effects of degradation of visual stimuli on components of the event-related potentials (ERP) in Go/Nogo reaction tasks. *Biol Psychol* 1986; **23**:21–38.
- Kopp B, Mattler U, Goertz R, Rist F. N2, P3 and the lateralized readiness potential in a nogo task involving selective response priming. *Electroencephalogr Clin Neurophysiol* 1996; **99**:19–27.
- Falkenstein M, Hoormann J, Hohnsbein J. ERP component in Go/Nogo tasks and their relation to inhibition. *Acta Psychol* 1999; **101**:267–291.
- Sasaki K, Gemba H. Electrical activity in the prefrontal cortex specific to no-go reaction of condition hand movement with colour discrimination in the monkey. *Exp Brain Res* 1986; **64**:603–606.
- Sasaki K, Gemba H, Tsujimoto T. Suppression of visually initiated hand movement by stimulation of the prefrontal cortex in the monkey. *Brain Res* 1989; **495**:100–107.
- Malloy P, Bihlre A, Duffy J. The orbitomedial frontal syndrome. *Arch Clin Neurophysiol* 1993; **8**:185–201.
- Rubia K, Russell T, Overmeyer S, Brammer MJ, Bullmore ET, Sharma T *et al.* Mapping motor inhibition: conjunctive brain activations across different versions of Go/Nogo and stop tasks. *Neuroimage* 2001; **13**: 250–261.
- Taylor DA. Time course of context effects. *J Exp Psychol: Gen* 1977; **106**:404–426.
- Geczy I, Czizler I, Balazs L. Effects of cue information on response production and inhibition measured by event-related potentials. *Acta Physiolog Hungar* 1999; **86**:37–44.

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